Echelon Form, Rank, and Solutions of $Ax = 0$ and $Bx = 0$

Problem

Reduce A and B to echelon form, to find their ranks. Which variables are free?

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$$

Find the special solutions to $Ax = 0$ and $Bx = 0$. Find all solutions.

Solution

Matrix A

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}$$

Subtract Row 1 from Row 3:

$$\to \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Subtract $2\times$ Row 2 from Row 1:

$$\to \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is echelon form.

Rank of A

There are 2 pivot rows, so

$$\boxed{\text{rank}(A) = 2.}$$

Free Variables

Pivot columns: 1 and 2. Free variables: x_3, x_4.

Solve $Ax = 0$

System:

$$x_1 - 2x_3 + x_4 = 0$$
$$x_2 + x_3 = 0$$

Let

$$x_3 = s, \quad x_4 = t.$$

Then

$$x_2 = -s,$$
$$x_1 = 2s - t.$$

So

$$x = \begin{bmatrix} 2s - t \\ -s \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Special Solutions

$$\boxed{\begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}}$$

All solutions are linear combinations of these.

Matrix B

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Subtract $4\times$ Row 1 from Row 2:

$$R_2 \to R_2 - 4R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{bmatrix}$$

Subtract $7\times$ Row 1 from Row 3:

$$R_3 \to R_3 - 7R_1 \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

Now eliminate in Row 3:

$$R_3 \to R_3 - 2R_2 \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of B

There are 2 pivot rows, so

$$\boxed{\text{rank}(B) = 2.}$$

Free Variable

Pivot columns: 1 and 2. Free variable: x_3.

Solve $Bx = 0$

From row 2:

$$-3x_2 - 6x_3 = 0 \Rightarrow x_2 = -2x_3.$$

From row 1:

$$x_1 + 2x_2 + 3x_3 = 0.$$

Substitute:

$$x_1 + 2(-2x_3) + 3x_3 = 0 \Rightarrow x_1 - 4x_3 + 3x_3 = 0 \Rightarrow x_1 = x_3.$$

Let $x_3 = t$.
Then

$$x = \begin{bmatrix} t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}.$$

Special Solution

$$\boxed{\begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}}$$

All Solutions

$$\boxed{x = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad t \in R.}$$